$$\frac{\text{input image } (32 \times 32 \times 3)}{\text{patch size } (4 \times 4)} = \begin{array}{l}\text{number of patches/}\\\text{tokens per image}\end{array} \overset{(32\times32)/(4\times4)}{=} 64 \text{ patches}$$

patch embedding

embedding dimension: 64

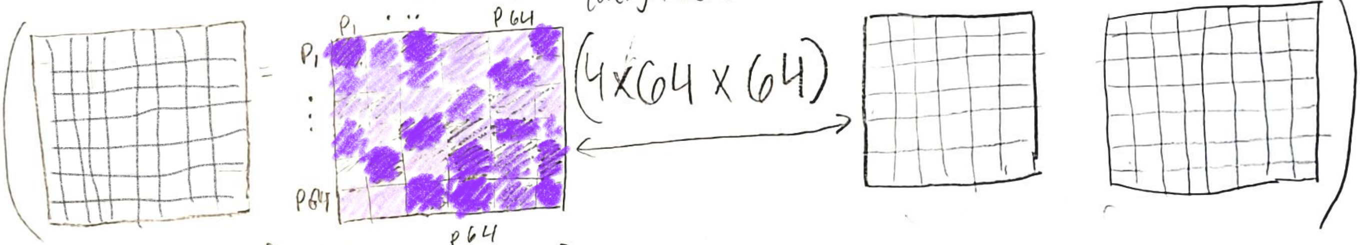
(hidden_dim: 128)

after PatchEmbedding: $[\underbrace{1}_{\text{batch}}, \underbrace{64}_{\text{num patches}}, \underbrace{64}_{\text{embed\_dim}}]$

Self attention on each token/patch

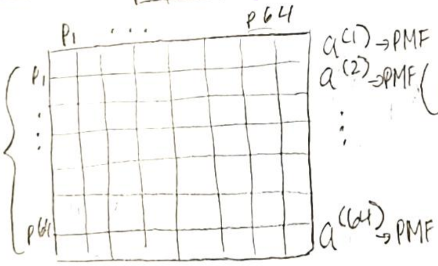
attention weights = $[\underbrace{1}_{\text{batch}}, \underbrace{4}_{\text{num-heads}}, \underbrace{64}_{\text{query-tokens}}, \underbrace{64}_{\text{key-tokens}}]$ $= \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{\text{hidden\_dim}}}\right) \cdot V$

mean



=

mean attention weights



(64×64)

$$= \text{entropy} \begin{bmatrix} \sum a^{(1)} \\ \sum a^{(2)} \\ \vdots \\ \sum a^{(64)} \end{bmatrix}$$

interpreting attention weights as probability distribution over input token/patches

→ each one is a seperate distribution

$\boxed{\text{high entropy}}$
= more spread of attention across the image for that patch

$\boxed{\text{low entropy}}$
= focused attention in regions of the image for that patch